

overlap-calculator

User Manual

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overlap-calculator is a Python package for reproducible batch analysis of the spectral overlap between a molecular absorption spectrum and a reference light source. It treats two input families as equal first-class citizens in the same run:

- **Theoretical** — Gaussian TD-DFT `.out` / `.log` files.
- **Experimental** — tabular spectra as **CSV** (`.csv`), **Excel workbook** (`.xlsx`), or **legacy Excel** (`.xls`), with multi-sheet and multi-series selection.

Every TD-DFT spectrum is broadened with both Gaussian and Lorentzian line shapes, and the tool reports per-light-source absorbed-flux, absorbed-fraction, and shape-overlap metrics together with publication-grade TIFF plots.

This manual covers installation, the CLI, the HTTP API, the input and output schemas, the physical assumptions and exact formulas, and environment configuration. For worked examples on real data, see the `case_studies/` directory of the repository.

Project home: <https://github.com/pinarsyt/overlap-calculator>

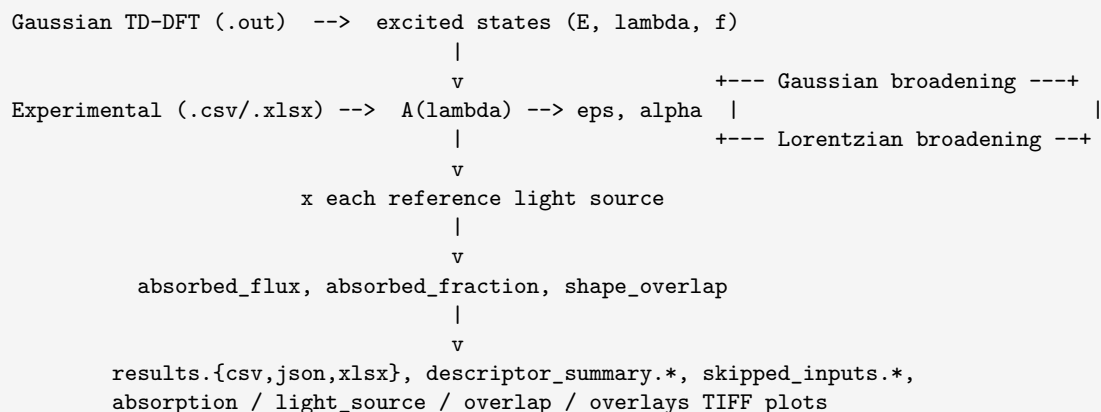
Archived release (all versions): <https://doi.org/10.5281/zenodo.19944515>

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1 Overview and Workflow



Pipeline stages:

1. **Input resolution** — A JSON manifest can be supplied directly or auto-generated from a directory of raw files. Experimental inputs expand to one row per numeric series column.
2. **Parsing** — Regex-based TD-DFT parser for .out files; pandas-based wavelength / signal loader for tabular spectra.
3. **Spectrum reconstruction** — Molar extinction $\varepsilon(\lambda)$ from TD-DFT states via oscillator-strength weighting on a shared wavelength grid; experimental signals are treated as absorbance directly.
4. **Beer–Lambert** — $\alpha(\lambda) = 1 - 10^{-A(\lambda)}$ with configurable reference concentration and path length.
5. **Overlap metrics** — Per (sample, light source, broadening) triplet.
6. **Export** — Tabular (CSV / JSON / XLSX), descriptor summary, skip report, and TIFF plots.

2 Installation

Requirements

- Python **3.12.x**
- Optional: Conda (recommended) or Docker

Conda (recommended)

```

conda env create -f environment.yml
conda activate overlap-calculator
pip install -e .

```

venv / pip

```

python -m venv .venv
source .venv/bin/activate          # Linux / macOS

```

```
# Windows PowerShell: .venv\Scripts\Activate.ps1
# Windows CMD:       .venv\Scripts\activate.bat
pip install -e .
```

Docker

The Docker image exposes the Flask API on port 8000 *inside* the container. To keep it from clashing with a native Flask instance on the host (which uses port 8000 directly), publish it on host port 8080:

```
docker build -t overlap-calculator .
docker run --rm -p 8080:8000 overlap-calculator
curl http://localhost:8080/health
```

Docker Compose

For a managed runtime (restart policy, healthcheck, pre-populated `GAUSS_*` environment), use the bundled compose file. It uses the same **host port 8080** mapping as `docker run` above, so the native instance on 8000 and either docker variant on 8080 can run side by side:

```
docker compose up -d          # starts the `overlap-calculator` service
curl http://localhost:8080/health
docker compose down
```

The compose service uses a Python-stdlib healthcheck against `/health` and restarts unless explicitly stopped. All runtime settings in §11 can be overridden via the `environment:` block or an adjacent `.env` file.

Verify the install

```
overlap-calculator version
overlap-calculator --help
```

3 Quick Start

Drop any combination of the supported input files into `input/files/` — TD-DFT outputs, CSV spectra, and Excel workbooks can live side by side in the same directory:

```
input/files/
|-- slurm-5473089.out      # Gaussian TD-DFT output (theoretical)
|-- dye_panel.xlsx        # Excel workbook, one or more sheets (experimental)
`-- uv_vis_run3.csv       # CSV spectrum (experimental)
```

Then run the two-step pipeline:

```
overlap-calculator generate-input --files-dir input/files --out input/input.json
overlap-calculator analyze --input input/input.json --out output
```

Results land under `output/tables/` and `output/plots/` (see §7).

4 Command-Line Interface

The package installs a single entry point, `overlap-calculator`, with three subcommands.

4.1 generate-input

```
overlap-calculator generate-input --files-dir PATH [--out PATH] [--log-level LEVEL] [--log-
  ↪ format FORMAT]
```

Flag	Default	Description
<code>--files-dir</code>	<code>input/files</code>	Directory containing TD-DFT and experimental files
<code>--out</code>	<code>input/input.json</code>	Output manifest path
<code>--log-level</code>	<code>INFO</code>	<code>DEBUG</code> , <code>INFO</code> , <code>WARNING</code> , <code>ERROR</code> , <code>CRITICAL</code>
<code>--log-format</code>	<code>text</code>	<code>text</code> or <code>json</code>

4.2 analyze

```
overlap-calculator analyze --input PATH [--out PATH] [--sigma-ev FLOAT] [--wl-min FLOAT] [--
  ↪ wl-max FLOAT] [--num-points INT] [--concentration-m FLOAT] [--path-cm FLOAT] [--prefactor-
  ↪ mode {constant,frequency-resolved}] [--sigma-mode {fixed,marcus-hush}] [--reorganization-
  ↪ ev FLOAT] [--temperature-k FLOAT] [--calibration PATH] [--default-light-sources STR] [--
  ↪ light-source-file PATH]... [--plot-outputs / --no-plot-outputs] [--plot-dpi INT] [--
  ↪ ranking-outputs / --no-ranking-outputs] [--log-level LEVEL] [--log-format FORMAT]
```

Flag	Default	Description
<code>--input</code>	<code>input/input.json</code>	Manifest JSON (see §6)
<code>--out</code>	<code>output</code>	Output directory (created if absent)
<code>--sigma-ev</code>	<code>0.30</code>	Broadening standard deviation, eV
<code>--wl-min</code>	<code>200.0</code>	Lower bound of the wavelength grid, nm
<code>--wl-max</code>	<code>800.0</code>	Upper bound of the wavelength grid, nm
<code>--num-points</code>	<code>10000</code>	Number of grid points
<code>--concentration-m</code>	<code>1e-5</code>	Beer–Lambert reference concentration, mol L ⁻¹
<code>--path-cm</code>	<code>1.0</code>	Beer–Lambert reference path length, cm
<code>--prefactor-mode</code>	<code>constant</code>	<code>constant</code> (integrated intensity; default) or <code>frequency-resolved</code> (keeps the wavenumber factor inside the integrand; relevant for NIR bands). See §9.6.
<code>--sigma-mode</code>	<code>fixed</code>	<code>fixed</code> (use <code>--sigma-ev</code>) or <code>marcus-hush</code> ($\sigma = \sqrt{2\lambda k_B T}$ from <code>--reorganization-ev</code> and <code>--temperature-k</code>). See §9.6.

Flag	Default	Description
<code>--reorganization-ev</code>	0.30	Reorganization energy λ , eV; used when <code>--sigma-mode marcus-hush</code>
<code>--temperature-k</code>	298.15	Temperature in K for the Marcus-Hush width
<code>--calibration</code>	<i>none</i>	Optional TOML calibration file (CLI and library only). See §9.7.
<code>--default-light-sources</code>	AM15G,LEDB4,...	Comma-separated bundled-source list
<code>--light-source-file</code>	<i>none</i>	Repeatable path to a custom light-source CSV
<code>--plot-outputs</code>	on	Emit TIFF plots (<code>--no-plot-outputs</code> to disable)
<code>--plot-dpi</code>	400	TIFF plot resolution; use 600 for larger publication exports
<code>--ranking-outputs</code>	on	Emit grouped ranking tables and bar charts (gaussian/lorentzian $\times$ <code>absorbed_fraction / shape_overlap</code>); disable with <code>--no-ranking-outputs</code>
<code>--log-level</code>	INFO	Logging verbosity
<code>--log-format</code>	text	text or json

4.3 version

Prints the installed package version.

```
overlap-calculator version
```

5 HTTP API

The examples in this section use port 8000, which is the **native Flask** port (started with `python -m overlap_calculator.api.app`). If you run the same API under **Docker** (either `docker run -p 8080:8000` or `docker compose up`), use port 8080 instead — the request bodies are otherwise identical.

5.1 Endpoints

Method	Path	Purpose
GET	<code>/health</code>	Liveness probe — returns <code>{"status":"ok"}</code>
POST	<code>/analyze</code>	Multipart upload; returns a ZIP of results

POST `/analyze` accepts the following form fields:

Form field	Type	Description
<code>files</code>	<code>file[]</code> (req.)	Mixed <code>.out/.csv/.xlsx/.xls</code> inputs
<code>light_source_files</code>	<code>file[]</code> (opt.)	Custom light-source CSVs

Form field	Type	Description
light_source_csv	file (opt.)	Back-compat alias for a single custom CSV
sigma_ev	float (opt.)	Broadening σ , eV (default 0.30)
wl_min, wl_max	float (opt.)	Wavelength window bounds, nm
num_points	int (opt.)	Grid size (default 10000)
concentration_m	float (opt.)	Beer–Lambert reference concentration (default 1e-5)
path_cm	float (opt.)	Beer–Lambert reference path length (default 1.0)
prefactor_mode	str (opt.)	constant (default) or frequency-resolved
sigma_mode	str (opt.)	fixed (default) or marcus-hush
reorganization_ev	float (opt.)	Reorganization energy λ , eV; used when sigma_mode=marcus-hush (default 0.30)
temperature_k	float (opt.)	Temperature in K for the Marcus–Hush width (default 298.15)
default_light_sources	str (opt.)	Comma-separated bundled-source list
plot_outputs	bool (opt.)	Enable / disable plot generation
plot_dpi	int (opt.)	TIFF plot resolution (default 400; use 600 for larger exports)
ranking_outputs	bool (opt.)	Enable / disable grouped ranking tables and bar charts (default true)
sheet_overrides	str (opt.)	JSON object {"<original-filename>": "<sheet-name>"} forcing a specific sheet for uploaded .xlsx/.xls workbooks

Response: **application/zip** containing every file **overlap-calculator analyze** would have written. Uploads are processed in a per-request workspace and are not persisted on the server.

5.2 Example

```
curl -X POST http://localhost:8000/analyze \
-F "files=@input/files/theoretical.out" \
-F "files=@input/files/experimental.xlsx" \
-F "sigma_ev=0.25" \
-F "plot_dpi=400" \
-F "ranking_outputs=true" \
-F "default_light_sources=AM15G,LEDB4" \
--output analysis_outputs.zip
```

To force a specific sheet in a multi-sheet workbook, add a **sheet_overrides** form field whose value is a JSON object keyed by the original filename:

```
curl -X POST http://localhost:8000/analyze \
-F "files=@input/files/dye_panel.xlsx" \
-F 'sheet_overrides={"dye_panel.xlsx": "raw"}' \
--output analysis_outputs.zip
```

Filenames that do not appear in the map use the same auto-discovery as **generate-input**: the first sheet with a wavelength column and at least one numeric signal column is selected.

A Postman collection is provided at `docs/postman_collection.json`.

6 Input Format

6.1 Accepted input files

`overlap-calculator` accepts two input families in the same run. Both are first-class: a single manifest can mix them freely, and the pipeline produces identical downstream columns, plots, and overlap metrics.

Family	Extensions	Parser	Notes
Theoretical	<code>.out</code> , <code>.log</code>	Regex-based Gaussian TD-DFT parser	File must contain a TD route and Excited State lines
Experimental	<code>.csv</code>	<code>pandas.read_csv</code>	One wavelength column and one or more numeric signal columns
Experimental	<code>.xlsx</code> , <code>.xls</code>	<code>pandas.read_excel</code> (openpyxl/xlrd)	Multi-sheet workbooks supported via <code>sheet_name</code> ; generate-input auto-selects the first analyzable sheet

Bulk uploads via the HTTP API share the same matrix (see `GAUSS_UPLOAD_ALLOWED_EXTS` in §11).

6.2 Manifest (`input.json`)

```
[
  {
    "input_type": "theoretical",
    "sample_id": "B1",
    "source_path": "files/slurm-5473089.out",
    "series_name": null,
    "sheet_name": null
  },
  {
    "input_type": "experimental",
    "sample_id": "channel_A",
    "source_path": "files/dye_panel.xlsx",
    "series_name": "channel_A",
    "sheet_name": "raw"
  }
]
```

Field	Req?	Description
<code>input_type</code>	yes	"theoretical" (TD-DFT <code>.out</code>) or "experimental" (tabular spectrum)
<code>sample_id</code>	yes	Unique sample identifier; used in filenames and result rows

Field	Req?	Description
<code>source_path</code>	yes	Path to the raw file; resolved relative to the manifest, then to the CWD
<code>series_name</code>	exp.	Column name of the numeric series in a tabular file
<code>sheet_name</code>	exp.	Sheet name for <code>.xlsx/.xls</code> inputs; in hand-written manifests, omit or <code>null</code> to read the first workbook sheet

`generate-input` fills these automatically. The `sample_id` is derived as follows:

- **Theoretical** (`.out`) — if the Gaussian route contains a `%chk=<name>.chk` line, the checkpoint stem is used (e.g. `%chk=B1_td.chk` $\Rightarrow$ `sample_id` = "B1_td"). This keeps human-meaningful molecule labels even when the raw files are queue-system artefacts such as `slurm-5473089.out`. If no `%chk` line is present, the file stem is used as a fallback.
- **Experimental** — one manifest entry is produced per numeric series column, with `sample_id` = "`<series_name>`" (the series column name itself, so `Coumarin_1` rather than a long file-stem prefix).

If a manifest is supplied by hand and `sample_id` is omitted, the same resolution rule is applied at analysis time.

6.3 Experimental data template

The same column layout applies to CSV, XLSX, and XLS files.

CSV (`.csv`)

```
wavelength_nm,sample_A,sample_B
300,0.12,0.08
310,0.15,0.10
320,0.18,0.14
```

Excel workbook (`.xlsx/.xls`). The first row is interpreted as the header. Any numeric columns beside the wavelength axis are treated as independent samples:

wavelength_nm	channel_A	channel_B	channel_C
300	0.12	0.08	0.05
310	0.15	0.10	0.07
320	0.18	0.14	0.09

Multi-sheet workbooks are fully supported. During `generate-input`, if no sheet override is supplied, the workbook sheets are scanned in order and the first sheet with a wavelength column and at least one numeric signal column is selected. In a hand-written manifest, if `sheet_name` is omitted from an entry (or set to `null`), analysis reads the first workbook sheet. To force a specific sheet, edit the generated manifest entry or author it by hand:

```
{
  "input_type": "experimental",
  "sample_id": "dye_panel__channel_A",
  "source_path": "files/dye_panel.xlsx",
  "series_name": "channel_A",
  "sheet_name": "raw"
}
```

Shared requirements (both formats).

- Exactly one wavelength column; accepted aliases (case-insensitive): `wavelength`, `wavelength_nm`, `lambda`, `wl`, `nm`. If no named wavelength column is found, the first numeric column is used.
- One or more numeric signal columns. Each signal is interpreted as absorbance $A(\lambda)$ and fed directly to the Beer–Lambert absorptance definition (§9).
- Rows with non-numeric or missing values in either the wavelength or signal column are dropped silently; a series must retain at least 10 valid rows to be analysed.

6.4 Custom light sources

A custom light source is a two-column CSV with one wavelength column and a single numeric intensity column. The filename stem becomes the `light_source_name` in the results table.

7 Output Schema

```
output/
|-- run_manifest.json           % provenance record
|-- tables/
|   |-- results.{csv,json,xlsx}
|   |-- results_timings.{csv,json,xlsx}
|   |-- descriptor_summary.{csv,xlsx}
|   |-- ranking_by_light_source__<metric>.{csv,json,xlsx}    % 4 metric variants
|   |-- ranking_by_sample__<metric>.{csv,json,xlsx}          % 4 metric variants
|   |-- skipped_inputs.{csv,json}          % emitted only if any inputs were skipped
`-- plots/
    |-- absorption/<sample>__{absolute,normalized}.tiff
    |-- light_source/<light>.tiff
    |-- overlap/<sample>__<light>__combined__{absolute,normalized}.tiff
    |-- overlap/<sample>__<light>__<method>__{absolute,normalized}.tiff
    |-- overlays/absorptance_overlay__<method>__{absolute,normalized}.tiff
    |-- ranking/by_light_source/<metric>/<light>.tiff          % bar chart per light source
    |-- ranking/by_sample/<metric>/<sample>.tiff               % bar chart per sample
```

`<metric>` cycles through the four overlap descriptors: `gaussian_absorbed_fraction`, `lorentzian_absorbed_fraction`, `gaussian_shape_overlap`, `lorentzian_shape_overlap`. The ranking tables and bar charts can be suppressed with `--no-ranking-outputs` (CLI) or `rankin g_outputs=false` (API).

7.1 results.* columns

One row per (sample_id, light_source_name) pair. Shared columns:

Column	Description
source_type	theoretical or experimental
sample_id	Unique sample identifier
light_source_name	AM15G, LEDB1, LEDB2, LEDB3, LEDB4, CIEFL10, or custom stem
sigma_ev	Broadening σ used, eV (theoretical inputs)
light_flux_total	$\int I(\lambda) d\lambda$ over the working grid
reference_concentration_molar	Beer–Lambert c used, mol L^{-1}
reference_path_cm	Beer–Lambert L used, cm

Column	Description
absorption_unit	Unit of ε / A in the exported row
light_source_unit	Unit of $I(\lambda)$ (e.g. $\text{W m}^{-2} \text{nm}^{-1}$ for AM1.5G)
absorbed_flux_unit	Unit of the absorbed-flux integral
td_method_basis	Route metadata from the TD-DFT file (null for experimental)
td_solvent	PCM solvent from the TD-DFT file (null for experimental)
prefactor_mode	constant or frequency-resolved — the prefactor convention used for this row
sigma_mode	fixed or marcus-hush — the width-source used for this row
software_version	Package version string at run time

Per-broadening columns (prefixes `gaussian_` and `lorentzian_`):

Suffix	Meaning
lambda_max_nm	Peak wavelength of the broadened absorptance
molar_absorptivity_max	Peak ε in $\text{M}^{-1} \text{cm}^{-1}$ (theoretical only)
absorptance_max	Peak Beer–Lambert $\alpha(\lambda)$
absorbed_flux	$\int \alpha(\lambda) I(\lambda) d\lambda$
absorbed_fraction	$\text{absorbed_flux} / \text{light_flux_total} \in [0, 1]$
shape_overlap	Dimensionless shape comparator on max-normalised spectra

Broadening deltas (Gaussian – Lorentzian): `delta_absorbed_flux`, `delta_absorbed_fraction`, `delta_shape_overlap`, `abs_delta_absorbed_fraction`.

7.2 descriptor_summary.*

A condensed table of the most actionable per-sample descriptors across all light sources, suitable as a starting point for ranking.

7.3 ranking_by_light_source__<metric>.* and ranking_by_sample__<metric>.*

Two complementary grouped rankings answer the two practical questions that arise from a multi-sample, multi-light-source run:

- `ranking_by_light_source__<metric>.{csv,json,xlsx}` — within each light source, samples are ordered from the highest to the lowest value of the chosen overlap metric. Use this to ask: *“Given this light source, which molecule absorbs (or matches in shape) best?”*
- `ranking_by_sample__<metric>.{csv,json,xlsx}` — within each sample, light sources are ordered the same way. Use this to ask: *“For this molecule, which light source gives the strongest overlap?”*

Each table is emitted four times — once per `<metric>` — covering both broadening methods (`gaussian_*`, `lorentzian_*`) and both metric families (`absorbed_fraction`, `shape_overlap`):

<metric>	Sorts by
<code>gaussian_absorbed_fraction</code>	Beer–Lambert absorbed fraction with Gaussian broadening; intensity-weighted

<metric>	Sorts by
lorentzian_absorbed_fraction	Beer–Lambert absorbed fraction with Lorentzian broadening; intensity-weighted
gaussian_shape_overlap	Shape-only correlation (intensity-independent) under Gaussian broadening
lorentzian_shape_overlap	Shape-only correlation (intensity-independent) under Lorentzian broadening

Both interpretations are reported because they answer different scientific questions: **absorbed_fraction** tracks the fraction of incident photons that are actually absorbed under the chosen Beer–Lambert reference (c, L) and is the right figure of merit for photocatalysis, photovoltaic action, or photo-excitation yield; **shape_overlap** is a brightness-independent spectral matching score, appropriate when comparing differently-scaled illuminants on equal footing (§9.4).

Each row carries a **rank** column (1 = best within the group) and all four metric columns side by side, so the file can be re-sorted client-side without rerunning the pipeline. Companion bar charts are written to `plots/ranking/by_light_source/<metric>/<light>.tiff` and `plots/ranking/by_sample/<metric>/<sample>.tiff`; titles and axis labels embed the chosen metric and the Beer–Lambert reference conditions to keep the figures unambiguous.

7.4 skipped_inputs.*

Emitted whenever at least one input is rejected. The `tables/skipped_inputs.{csv,json}` rows carry **sample_id** and **reason** columns. The manifest-side `input/skipped_inputs.json` produced by `generate-input` additionally carries the resolved **source_path**. Each **reason** string is prefixed with the stage at which the input was skipped (`INPUT_SKIPPED` / `INPUT_ERROR` / `ANALYSIS_ERROR`) followed by the error message.

7.5 results_timings.*

One row per (**sample_id**, **light_source_name**) pair. Shared columns: **source_type**, **sample_id**, **light_source_name**, **plot_ms**. Each broadening method contributes a five-column block (**gaussian_parse_tddft_ms**, **gaussian_broadening_ms**, **gaussian_light_source_ms**, **gaussian_integrals_ms**, **gaussian_total_sample_ms**, and the matching **lorentzian_*** quintuplet). The row ends with **total_pair_ms** (sum of both methods' **total_sample_ms**) and **total_with_plot_ms** (**total_pair_ms**+**plot_ms**).

7.6 Plots

TIFF plots are emitted at 400 dpi by default and include both absolute and max-normalised variants. Use `--plot-dpi600` on the CLI or `plot_dpi=600` in the API when larger publication-resolution files are worth the additional disk and transfer size.

Directory	Content
<code>absorption/</code>	Per-sample absorptance spectra, both broadenings overlaid
<code>light_source/</code>	Each reference light source in isolation
<code>overlap/<combined></code>	(sample, light) overlap — both broadenings on one axes
<code>overlap/<method></code>	(sample, light) overlap — single broadening
<code>overlays/</code>	Multi-sample absorptance overlays per broadening
<code>ranking/by_light_source/<metric>/</code>	Bar chart per light source ranking samples by <metric> (descending)

Directory	Content
ranking/by_sample/<metric>/	Bar chart per sample ranking light sources by <metric> (descending)

The overlap plots shade the geometric overlap area $\min(\alpha(\lambda), \hat{I}(\lambda))$ (absolute panel) and $\min(\hat{\alpha}(\lambda), \hat{I}(\lambda))$ (normalised panel). In the normalised panel the shaded region is exactly the integrand of `shape_overlap`, so the figure and the numerical metric report the same quantity: the area bounded above by the lower of the two curves at every wavelength.

7.7 run_manifest.json

Every `analyze` run writes `run_manifest.json` directly inside the output root directory (at the same level as `tables/` and `plots/`). The file records enough information to trace any result back to its exact inputs and parameters.

Top-level fields:

Field	Description
<code>software_version</code>	Installed package version string
<code>git_commit</code>	Full git SHA of the running commit; <code>null</code> if unavailable
<code>generated_at_utc</code>	ISO-8601 UTC timestamp of the run
<code>parameters</code>	Full resolved runtime parameter set: <code>sigma_ev</code> , <code>prefactor_mode</code> , <code>sigma_mode</code> , <code>reorganization_ev</code> , <code>temperature_k</code> , wavelength grid, Beer–Lambert constants, light-source lists, and the full calibration block when <code>--calibration</code> was supplied
<code>light_sources</code>	Sorted list of light-source names used in the run
<code>inputs</code>	Array — one object per source file (see below)
<code>results_count</code>	Number of <code>RunResult</code> rows produced
<code>skipped_count</code>	Number of inputs that were skipped

Each element of `inputs`:

Field	Description
<code>sample_id</code>	Sample identifier from the manifest
<code>input_type</code>	"theoretical" or "experimental"
<code>source_path</code>	Run-directory-relative path to the source file (never an absolute machine path); content fingerprinted by <code>sha256</code>
<code>series_name</code>	Series column name (experimental inputs only; otherwise <code>null</code>)
<code>sheet_name</code>	Sheet name for Excel inputs with an explicit sheet override; otherwise <code>null</code>
<code>sha256</code>	SHA-256 hex digest of the file contents at run time; <code>null</code> if the file was not readable

8 Bundled Light Sources

Name	Source	Unit
AM15G	NREL ASTM G-173 AM1.5G reference solar spectrum	$\text{W m}^{-2} \text{nm}^{-1}$

Name	Source	Unit
LEDB1	CIE 15 illuminant LED-B1 (opt-in; not in default set)	relative (shape only)
LEDB2	CIE 15 illuminant LED-B2	relative (shape only)
LEDB3	CIE 15 illuminant LED-B3	relative
LEDB4	CIE 15 illuminant LED-B4	relative
CIEFL10	CIE 15 fluorescent illuminant FL10	relative

Unit policy: because **AM15G** carries an absolute spectral irradiance, **absorbed_flux** for AM1.5G has units of W m^{-2} ; for the relative CIE LED / FL sources **absorbed_flux_unit** is **relative_integral** and only **absorbed_fraction** and **shape_overlap** are directly comparable across dyes.

Third-party spectral data. The LEDB1, LEDB2, LEDB3, LEDB4, and CIEFL10 spectra are official **CIE** data sets derived from *CIE 015:2018 Colorimetry, 4th Edition* (International Commission on Illumination, <https://cie.co.at/>). The **AM15G** spectrum is the ASTM G-173-03 AM1.5 global reference distributed by the U.S. National Renewable Energy Laboratory (NREL). See §13 for full citations and BibTeX entries.

9 Methodology

9.1 Extinction spectrum from TD-DFT

Each excited state i contributes a normalised line profile in wavenumber space $\tilde{\nu} = 10^7/\lambda[\text{nm}]$ (cm^{-1}):

$$g_i(\tilde{\nu}) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(\tilde{\nu}-\tilde{\nu}_i)^2}{2\sigma^2}\right), \quad (1)$$

$$l_i(\tilde{\nu}) = \frac{1}{\pi} \frac{\gamma}{(\tilde{\nu} - \tilde{\nu}_i)^2 + \gamma^2}, \quad \gamma = \sqrt{2\ln 2} \sigma \approx 1.1774 \sigma, \quad (2)$$

with σ in cm^{-1} obtained from the CLI **--sigma-ev** via $\sigma_{\text{cm}^{-1}} = \sigma_{\text{eV}} \cdot 8065.544$. Here σ is the Gaussian standard deviation ($\text{FWHM}_G = 2\sqrt{2\ln 2} \sigma \approx 2.3548 \sigma$); the Lorentzian half-width-at-half-maximum γ is set to $\sqrt{2\ln 2} \sigma$ so that *both line shapes share the same FWHM* under a given **--sigma-ev** value. This removes the implicit width mismatch between the two broadening models and keeps the Gaussian–Lorentzian sensitivity comparison a pure line-shape probe; the **delta_*** columns therefore reflect tail behaviour rather than width difference. Both profiles are normalised in cm^{-1} : $\int p(\tilde{\nu}) d\tilde{\nu} = 1$. The molar extinction coefficient is then

$$\varepsilon(\lambda) = 2.315 \times 10^8 \sum_i f_i \text{profile}_i(\tilde{\nu}(\lambda)) \quad [\text{M}^{-1} \text{cm}^{-1}], \quad (3)$$

where the prefactor comes from the integrated-absorption / oscillator-strength relation $\int \varepsilon d\tilde{\nu} = 2.315 \times 10^8 \sum_i f_i$.

9.2 Beer–Lambert absorbance

$$A(\lambda) = \varepsilon(\lambda) c L, \quad \alpha(\lambda) = 1 - 10^{-A(\lambda)} \quad (4)$$

with defaults $c = 1 \times 10^{-5} \text{ M}$, $L = 1 \text{ cm}$ (overridable via **--concentration-m** / **--path-cm**). For experimental inputs the user-provided signal is treated as absorbance A directly and fed to the same α definition without the Beer–Lambert conversion.

9.3 Light-overlap metrics

$$\text{absorbed_flux} = \int \alpha(\lambda) I(\lambda) d\lambda, \quad (5)$$

$$\text{light_flux_total} = \int I(\lambda) d\lambda, \quad (6)$$

$$\text{absorbed_fraction} = \frac{\text{absorbed_flux}}{\text{light_flux_total}} \in [0, 1], \quad (7)$$

$$\text{shape_overlap} = \frac{\int \min(\hat{\alpha}(\lambda), \hat{I}(\lambda)) d\lambda}{\int \hat{I}(\lambda) d\lambda}, \quad (8)$$

where $\hat{\alpha}$ and $\hat{I}$ denote max-normalised shapes. **shape_overlap** is the classical geometric spectral-overlap area: at every wavelength the integrand follows whichever of the two normalised profiles is smaller, so the shaded region in a normalised overlap plot equals the numerator directly. This descriptor is independent of absolute strength and lets LED / FL sources (relative intensities) still be compared.

9.4 Choosing between **absorbed_fraction** and **shape_overlap**

The two scalar overlap descriptors answer different questions and are *both* emitted in the ranking tables and bar charts (§7.3) so that no single metric is forced on the user:

- **absorbed_fraction** is the right figure of merit when the application cares about the *number of photons (or the energy) actually absorbed* under a defined Beer–Lambert reference state (c, L) — for example, photocatalytic turnover under a given illuminant, photovoltaic action spectra, or photo-excitation yield. It folds in *both* spectral overlap and the absolute spectral intensity of the light source: a brighter source over the absorbing band is rewarded.
- **shape_overlap** is intensity-independent and answers the brightness-blind question “*how well does the light source’s spectral envelope match the molecule’s absorption envelope?*” It is the appropriate descriptor when the spectra under comparison carry incompatible intensity units (e.g. $\text{W m}^{-2} \text{ nm}^{-1}$ for AM1.5G versus **relative** for the CIE LED templates) and the reader needs a fair side-by-side ordering.

Each descriptor is reported under both Gaussian and Lorentzian broadening (§9), so the four ranking tables together expose the full sensitivity surface to the line-shape choice.

9.5 Skip policy

An input is diverted to **skipped_inputs.*** when

- the **.out** file lacks a TD route or any **Excited State** line (tagged **INPUT_SKIPPED:fileisnotaTD-DFToutput**);
- an experimental file cannot be read or has no numeric signal column (tagged **INPUT_SKIPPED:failedtoreadexperimentalfile** or **INPUT_SKIPPED:experimentalfilehasnonumericseries**);
- parsing fails at analysis time (tagged **ANALYSIS_ERROR:Failedtoparse...**).

9.6 Choice of prefactor and broadening width

Prefactor modes

The oscillator-strength prefactor $P = 2.315 \times 10^8 \text{ M}^{-1} \text{ cm}^{-2}$ comes from the Hilborn/Mulliken oscillator-strength relation: the integrated band area equals $P \cdot f$.

Constant prefactor (`--prefactor-mode constant, default`).

$$\varepsilon(\nu) = \sum_i P f_i g_i(\nu - \nu_i) \quad (9)$$

where g_i is a normalized lineshape. The band area equals $P f_i$ independent of ν_i . This is the validated convention for the 200–800 nm visible range.

Frequency-resolved prefactor (`--prefactor-mode frequency-resolved`).

$$\varepsilon(\nu) = \sum_i P f_i \frac{\nu}{\nu_i} g_i(\nu - \nu_i) \quad (10)$$

The integrated band area is unchanged for a symmetric lineshape. The two modes differ only in peak height (by approximately $+\frac{1}{2}(\sigma/\nu_i)^2$) and in a small blue-shift of the peak, of order $O((\sigma/\nu_i)^2)$. This difference is below $\sim 2\%$ over 200–800 nm for $\sigma = 0.30 \text{ eV}$, and grows into the NIR.

Recommendation: use `constant` for the validated visible range; use `frequency-resolved` for NIR bands.

Lineshape definitions

Gaussian:

$$g(x) = \frac{1}{\sigma_{\text{cm}} \sqrt{2\pi}} \exp\left(-\frac{x^2}{2\sigma_{\text{cm}}^2}\right), \quad \sigma_{\text{cm}} = \sigma_{\text{eV}} \times 8065.5439 \text{ cm}^{-1}. \quad (11)$$

Lorentzian (FWHM-matched to the Gaussian):

$$l(x) = \frac{\gamma/\pi}{x^2 + \gamma^2}, \quad \gamma = \sigma_{\text{cm}} \sqrt{2 \ln 2}. \quad (12)$$

Marcus–Hush width

The classical high-temperature Marcus absorption band is a Gaussian in energy whose variance is $2\lambda k_B T$, giving

$$\sigma_E = \sqrt{2\lambda k_B T}. \quad (13)$$

With $k_B T = 0.025693 \text{ eV}$ at 298.15 K, for $\lambda = 0.30 \text{ eV}$: $\sigma_E = 0.12416 \text{ eV}$ (FWHM = 0.29234 eV).

Caveat. This is the classical (high-temperature) limit. For organic vibrational baths ($\sim 1400 \text{ cm}^{-1}$), $\hbar\omega \gg k_B T$, so the Marcus–Hush width is a first-order estimate rather than a quantitative bandshape. For quantitative bandshape work, supply vibronic (Franck–Condon) spectra through the experimental/spreadsheet (CSV/XLSX) input branch. The tool is not locked to Gaussian broadening and processes any tabular absorption spectrum through the identical descriptor pipeline.

9.7 Calibration

A calibration block supplied via `--calibration PATH` (CLI and library) is applied *before* broadening and does not change the output path.

The calibration performs three operations, all optional and independent:

1. **Energy map** $E_{\text{cal}} = aE + b$ (E and b in eV). Wavelength is recomputed from the calibrated energy via $\lambda = 1239.841984 \text{ eV nm} / E_{\text{cal}}$ only when the map is non-identity (i.e. when $a \neq 1$ or $b \neq 0$).
2. **Oscillator-strength scaling** $f_{\text{cal}} = \alpha f$.
3. **Per-band overrides**: optional per-transition `sigma_ev` or `reorganization_ev` that take precedence over the global values.

Guards: $\alpha > 0$ and every $E_{\text{cal}} > 0$ are enforced. The identity calibration ($a = 1$, $b = 0$, $\alpha = 1$, no band overrides) produces output bit-identical to running without `--calibration`.

TOML schema

```
[energy]
a = 1.0
b = 0.0

[oscillator]
alpha = 1.0

[[band]]
index = 1          # 1-based excited-state index
sigma_ev = 0.25    # width override (eV)
reorganization_ev = 0.40
```

All sections and all keys are optional. Omitted keys take their identity or default values.

Note. `--calibration` is available on the CLI and via the Python library. The HTTP API does not accept calibration files.

10 Use as a Library

`overlap_calculator` is installable as a Python package and exposes a fully typed public API. Every name below is importable directly from `overlap_calculator`.

Minimal snippet

```
from overlap_calculator import analyze
from overlap_calculator.models import AnalysisInput

items = [AnalysisInput(
    input_type="theoretical",
    source_path="tddft.out",
    sample_id="dye-1",
)]
results, skipped = analyze(items, default_light_sources=["AM15G"])
```

analyze signature

```

analyze(
    items: list[AnalysisInput],
    *,
    sigma_ev: float = 0.30,
    wl_min: float = 200.0,
    wl_max: float = 800.0,
    num_points: int = 10000,
    concentration_molar: float = 1e-5,
    path_cm: float = 1.0,
    default_light_sources: list[str] | None = None,
    custom_light_source_paths: list[Path] | None = None,
    prefactor_mode: str = "constant",
    sigma_mode: str = "fixed",
    reorganization_ev: float = 0.30,
    temperature_k: float = 298.15,
    calibration: Calibration | None = None,
) -> tuple[list[RunResult], list[AnalysisSkip]]

```

Public API names

Name	Purpose
<code>analyze</code>	High-level entry point — full pipeline
<code>analyze_inputs</code>	Lower-level batch runner
<code>export_results</code>	Write tables and plots from a <code>results</code> list
<code>prepare_output_dir</code>	Create the output directory tree
<code>build_extinction_spectrum</code>	Reconstruct $\varepsilon(\lambda)$ from TD-DFT excited states
<code>compute_absorbance</code>	Beer–Lambert $A(\lambda) = \varepsilon c L$
<code>compute_absorptance</code>	$\alpha(\lambda) = 1 - 10^{-A}$
<code>shape_overlap</code>	Dimensionless spectral shape comparator
<code>integrate_light_flux</code>	$\int I(\lambda) d\lambda$
<code>integrate_absorbed_flux</code>	$\int \alpha(\lambda) I(\lambda) d\lambda$
<code>load_light_sources</code>	Load bundled and custom light sources
<code>marcus_hush_sigma_ev</code>	$\sigma = \sqrt{2\lambda k_B T}$ in eV
<code>Calibration</code>	Pydantic model for the calibration block
<code>load_calibration</code>	Parse a TOML calibration file
<code>apply_calibration</code>	Apply a calibration to a list of excited states
<code>build_run_manifest</code>	Assemble a provenance manifest dict
<code>write_run_manifest</code>	Write <code>run_manifest.json</code> to disk
<code>__version__</code>	Installed package version string

11 Configuration and Environment Variables

All settings are read via `pydantic-settings` with the `GAUSS_` prefix; a `.env` file in the working directory is honoured.

Variable	Default	Purpose
<code>GAUSS_LOG_LEVEL</code>	<code>INFO</code>	Logging verbosity
<code>GAUSS_LOG_FORMAT</code>	<code>text</code>	<code>text</code> or <code>json</code>
<code>GAUSS_PLOT_OUTPUTS</code>	<code>true</code>	Emit TIFF plots

Variable	Default	Purpose
GAUSS_PLOT_DPI	400	Default TIFF plot resolution
GAUSS_RANKING_OUTPUTS	true	Emit grouped ranking tables and bar charts
GAUSS_SIGMA_EV	0.30	Default broadening σ
GAUSS_WAVELENGTH_MIN_NM	200.0	Grid lower bound, nm
GAUSS_WAVELENGTH_MAX_NM	800.0	Grid upper bound, nm
GAUSS_WAVELENGTH_POINTS	10000	Number of grid points
GAUSS_REFERENCE_CONCENTRATION_MOLAR	1e-5	Beer–Lambert c
GAUSS_REFERENCE_PATH_CM	1.0	Beer–Lambert L
GAUSS_DEFAULT_LIGHT_SOURCES	AM15G,LEDB4,...	Default bundled-source list
GAUSS_UPLOAD_MAX_MB	50	API upload size cap
GAUSS_UPLOAD_ALLOWED_EXTS	.out,.log,.csv,.xlsx,.xlsx	Accepted upload extensions

CLI flags override environment variables when both are supplied.

12 Troubleshooting

No output generated

Inspect `output/tables/skipped_inputs.*` first and re-run with `--log-level DEBUG`. A common cause is a `.out` file that is an OPT job, not a TD-DFT job, so it contains no Excited State blocks.

ParseError on an experimental file

Confirm one column is a wavelength axis (alias list in §6) and at least one other column is numeric. Non-numeric / date-formatted columns are dropped silently.

Custom light source is ignored

Check that the file has exactly one wavelength column and one numeric intensity column. The filename stem becomes the `light_source_name`; if that name collides with a bundled source the custom file wins.

CLI options seem outdated

After switching branches, re-install the package: `pip install -e ..`

Unexpected absorbed_fraction near 0 for a CIE LED source

Check that the dye's $\lambda_{\max}$ falls within the LED emission band. `shape_overlap` is often more informative than `absorbed_fraction` for narrow-band relative sources.

run_manifest.json is missing

The manifest is written by `analyze`. It will not be present for runs that were aborted before completion. Re-run the pipeline to produce it.

git_commit is null in run_manifest.json

The package was installed from a wheel or archive rather than from a git checkout, so no commit SHA is available. The `software_version` field still identifies the release.

13 Citation and License

If you use `overlap-calculator` in published work, please cite the concept DOI below. It resolves to the latest archived release on Zenodo, so the citation stays valid across versions without needing an update.

Seyitdanlioğlu, P. *overlap-calculator: Batch spectral-overlap analysis between TD-DFT / experimental absorption and reference light sources*. Zenodo. <https://doi.org/10.5281/zenodo.19944515>

BibTeX:

```
@software{seyitdanlioglu_overlap_calculator,
  author    = {Seyitdanlioglu, P.},
  title     = {{overlap-calculator}: Batch spectral-overlap analysis
              between TD-DFT / experimental absorption and reference
              light sources},
  year      = {2026},
  publisher = {Zenodo},
  doi       = {10.5281/zenodo.19944515},
  url       = {https://doi.org/10.5281/zenodo.19944515}
}
```

Acknowledgement of third-party data

The bundled CIE LED illuminants (LEDB1, LEDB2, LEDB3, LEDB4) and fluorescent illuminant (CIEFL10) are official **CIE** data sets derived from *CIE 015:2018 Colorimetry, 4th Edition*. The AM15G spectrum is the ASTM G-173-03 AM1.5 global reference distributed by the U.S. National Renewable Energy Laboratory (NREL). The bundled experimental absorbance workbook (input/files/organic_uvvis_photochemcad_dataset.xlsx) is derived from the **PhotochemCAD** spectral database. If you use `overlap-calculator` in published work, please cite the data sets alongside the software:

CIE (2018). *Relative spectral power distributions of illuminants representing typical LED lamps, 1 nm spacing*. International Commission on Illumination, Vienna, AT. DOI: [10.25039/CIE.DS.dhew57sd](https://doi.org/10.25039/CIE.DS.dhew57sd)

CIE (2018). *Relative spectral power distributions of illuminants representing typical fluorescent lamps, 1 nm wavelength steps*. International Commission on Illumination, Vienna, AT. DOI: [10.25039/CIE.DS.54hy6srn](https://doi.org/10.25039/CIE.DS.54hy6srn)

CIE (2018). *CIE 015:2018 Colorimetry, 4th Edition*. International Commission on Illumination, Vienna, AT. <https://cie.co.at/publications/colorimetry-4th-edition/>

NREL. *Reference Air Mass 1.5 Spectra (ASTM G-173-03)*. U.S. National Renewable Energy Laboratory. <https://www.nrel.gov/grid/solar-resource/spectra-am1.5.html>

Taniguchi, M., & Lindsey, J. S. (2018). Database of absorption and fluorescence spectra of >300 common compounds for use in PhotochemCAD. *Photochemistry and Photobiology*, **94**(2), 290–327. DOI: [10.1111/php.12860](https://doi.org/10.1111/php.12860)

BibTeX:

```

@techreport{cie_led_illuminants_2018,
  author      = {{International Commission on Illumination}},
  title       = {Relative spectral power distributions of illuminants
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  institution = {CIE Central Bureau},
  address     = {Vienna, AT},
  year        = {2018},
  doi         = {10.25039/CIE.DS.dhbw57sd},
  url         = {https://doi.org/10.25039/CIE.DS.dhbw57sd}
}

@techreport{cie_fl_illuminants_2018,
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  title       = {Relative spectral power distributions of illuminants
                 representing typical fluorescent lamps, 1\,nm wavelength steps},
  institution = {CIE Central Bureau},
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  year        = {2018},
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  url         = {https://doi.org/10.25039/CIE.DS.54hy6srn}
}

@techreport{cie_015_2018,
  author      = {{International Commission on Illumination}},
  title       = {{CIE} 015:2018 Colorimetry, 4th Edition},
  institution = {CIE Central Bureau},
  address     = {Vienna, AT},
  year        = {2018},
  isbn        = {978-3-902842-13-8}
}

@misc{nrel_am15g,
  author      = {{National Renewable Energy Laboratory}},
  title       = {Reference Air Mass 1.5 Spectra ({ASTM} {G-173-03})},
  howpublished = {\url{https://www.nrel.gov/grid/solar-resource/spectra-am1.5.html}},
  institution = {U.S. Department of Energy}
}

@article{taniguchi_lindsey_photochemcad_2018,
  author      = {Taniguchi, Masahiko and Lindsey, Jonathan S.},
  title       = {Database of Absorption and Fluorescence Spectra of
                 {>}300 Common Compounds for Use in {PhotochemCAD}},
  journal     = {Photochemistry and Photobiology},
  volume      = {94},
  number      = {2},
  pages       = {290--327},
  year        = {2018},
  doi         = {10.1111/php.12860},
  url         = {https://doi.org/10.1111/php.12860}
}

```

Please consult the CIE at <https://cie.co.at/> for the authoritative reference tables and the latest errata.

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